

FOR MECHANICAL, ECE AND ROBOTICS TEAMS — A HARDWARE-LED PROJECT

A worked sample. Replace every word with your own project. Notice how much of the answer is carried by a figure rather than by a sentence.

A. PROJECT IDENTIFICATION

Project ID	PE-2026-ROB-014
Project Title	Autonomous Indoor Delivery Robot with Lift Integration
Domain	Robotics and Automation
Project Category	Integrated Hardware–Software
Department & Programme	Mechanical Engineering (lead), with ECE and CSE
Project Origin	Independent work, grown out of a Design Thinking mini project

B. TEAM DETAILS

#	Name of the Student	Roll Number	Programme / Year / Department	Mobile & E-mail
1	R. Aravindhan (Team Leader)	CH.EN.U4MEE22004	B.Tech Mechanical, III Year	9xxxxxxxxx
2	S. Meenakshi	CH.EN.U4ECE22031	B.Tech ECE, III Year	9xxxxxxxxx
3	K. Hariprasad	CH.EN.U4CSE23015	B.Tech CSE, II Year	9xxxxxxxxx

C. FACULTY MENTOR

Name & Designation	Dr. S. Ramkumar, Associate Professor
Department	Department of Mechanical Engineering
E-mail & Mobile	s_ramkumar@ch.amrita.edu · 9xxxxxxxxx

D. ABSTRACT AND KEYWORDS

Internal documents, laboratory samples and stores items are carried between blocks on this campus by hand, which ties up staff time and offers no record of what moved where. This project is a differential-drive robot that carries a locked 12 kg payload between floors without an escort. A 2-D LiDAR builds and holds a map of the corridors, an A* planner sets the route, and a local planner steers the robot around people and trolleys in real time. A relay module calls the lift over Bluetooth, so the robot moves between floors on its own. The cargo bay opens only against a one-time password sent to the recipient, which gives a delivery record. The chassis was built from aluminium extrusion with brushless drive motors and a 24 V lithium iron phosphate pack. Across 120 supervised runs on two floors the robot completed 94.2 per cent of deliveries without human help, docked within 3.1 centimetres of the target point, and ran for 6.4 hours on one charge.

Keywords: autonomous mobile robot; LiDAR SLAM; indoor navigation; lift integration; campus logistics

E. PROBLEM AND OBJECTIVES

Departments on this campus move documents, samples and small stores items between four blocks and three floors by hand. Two housekeeping staff spend an estimated 90 minutes a day on these trips, there is no record of what was handed to whom, and anything confidential travels in an open tray. Delivery robots sold for hospitals and offices cost upward of twelve lakh rupees and none of them talk to the lifts already installed here.

Objectives

- Carry a 12 kg payload between any two points on two floors without a human escort.
- Complete at least 90 per cent of delivery runs without human intervention.
- Dock within 5 centimetres of the marked drop point.
- Run for at least 6 hours on a single charge.
- Keep the total build cost under ₹1,20,000.

F. DESIGN AND TECHNICAL APPROACH

The robot is a differential-drive platform with two 200 mm brushless wheels on the axle line and a castor at the front, which keeps the turning circle inside a corridor width. A 2-D LiDAR on a rear mast builds the map and localises against it; a depth camera fills the LiDAR blind zone below knee height, where trolleys and feet are missed. Wheel encoders and an IMU feed an extended Kalman filter, so localisation survives a brief LiDAR occlusion in a crowded corridor. Global routes come from A* over the stored map and are followed by a dynamic-window local planner that reacts to people at up to 0.8 m/s. A Bluetooth relay module presses the lift call button, which avoided any modification to the lift controller and therefore any approval from the lift vendor. Everything runs as ROS 2 nodes on a single-board computer.

SYSTEM BLOCK DIAGRAM

SENSE → DECIDE → ACT

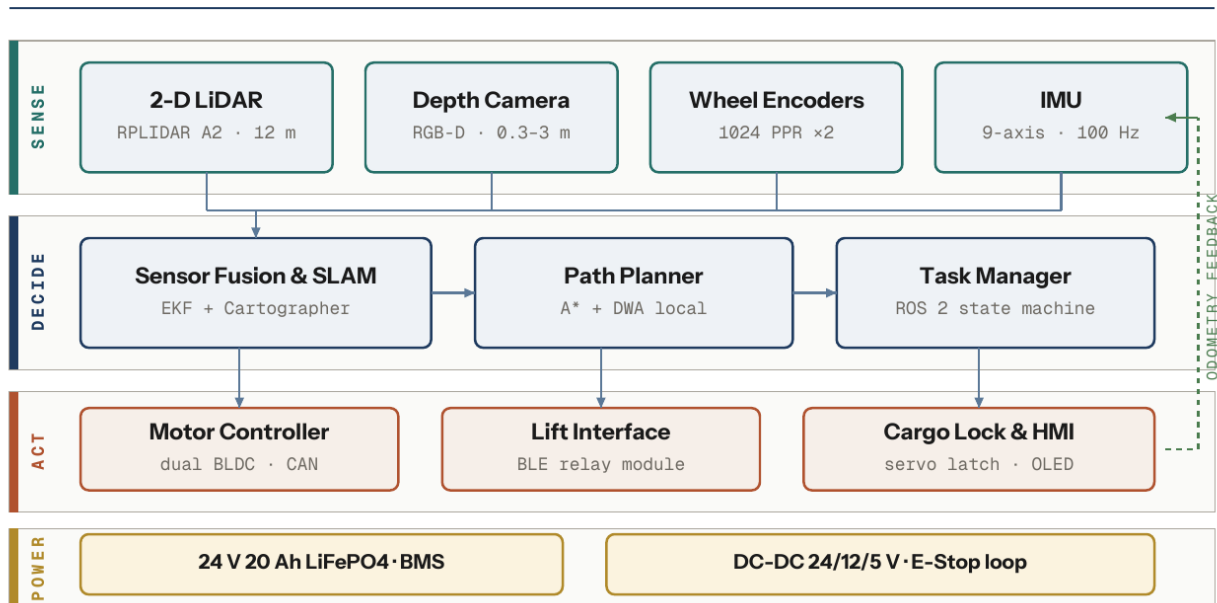


Figure 1 – Signal flow across the four subsystems. Dashed line is closed-loop odometry feedback.

CONCEPTUAL LAYOUT — SIDE ELEVATION

NOT TO SCALE

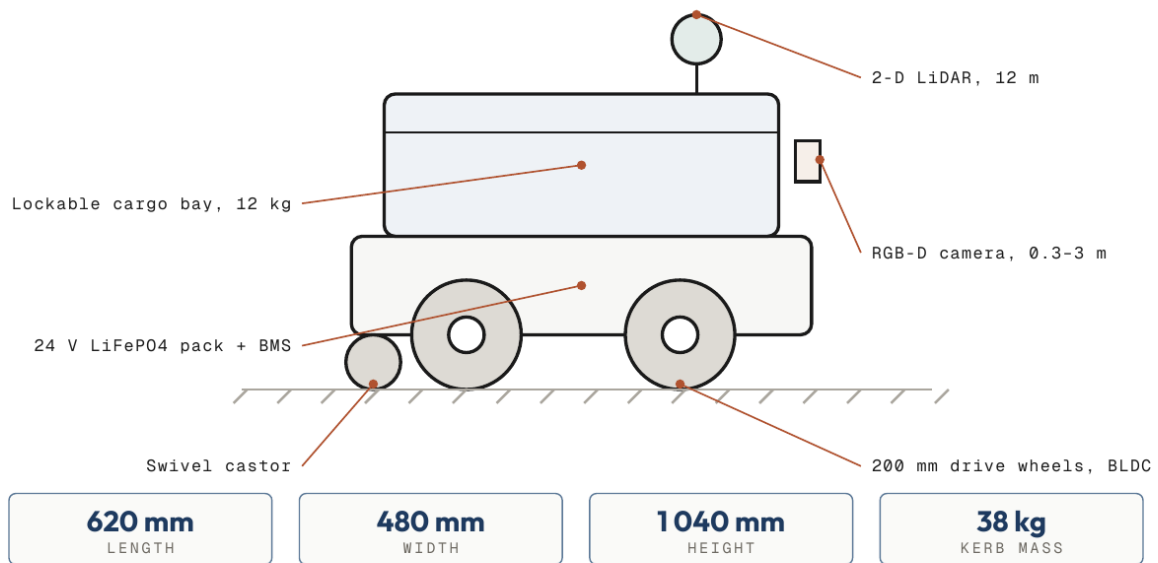
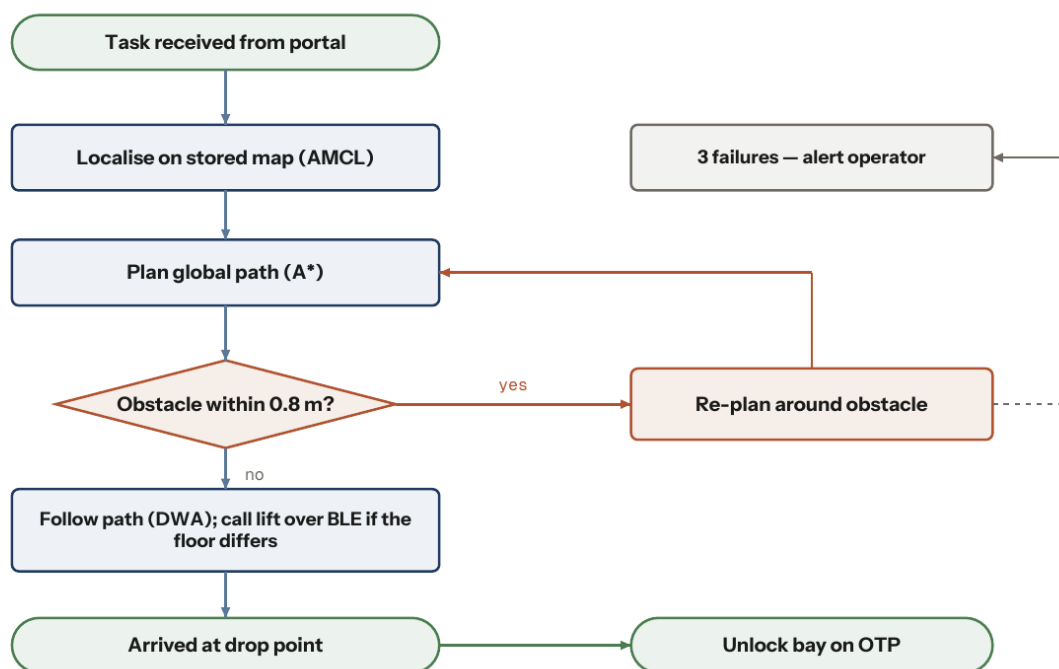


Figure 2 – Differential drive on the axle line, castor forward, sensor mast at the rear.

NAVIGATION AND DELIVERY LOGIC

ONE DELIVERY TASK



MEASURED: THE RE-PLAN LOOP IS ENTERED 2.4 TIMES PER RUN ON AVERAGE

Figure 3 – Decision logic for one delivery task, from portal request to cargo release.

Tools and technologies used: ROS 2 Humble, Python, C++, Cartographer SLAM, SolidWorks, Altium, NVIDIA Jetson Orin Nano, STM32 motor controller, RPLIDAR A2.

G. IMPLEMENTATION AND RESULTS

The chassis, drivetrain, castor and cargo body are fabricated and assembled. The 24 V power pack, battery management system and the emergency-stop loop are wired, tested and certified safe by the department workshop. LiDAR mapping and localisation are complete and the map of two floors is stored on board. Path

planning and obstacle avoidance work and are being tuned to reduce hesitation near glass walls. The Bluetooth lift interface calls the lift reliably but does not yet confirm which floor it arrived at, so that step is supervised at present. The cargo lock and the one-time-password screen are wired but the operator interface is still being written. Overall the robot is about 86 per cent complete and can be demonstrated end to end today, with the lift step supervised.

BUILD STATUS BY SUBSYSTEM

OVERALL 86%

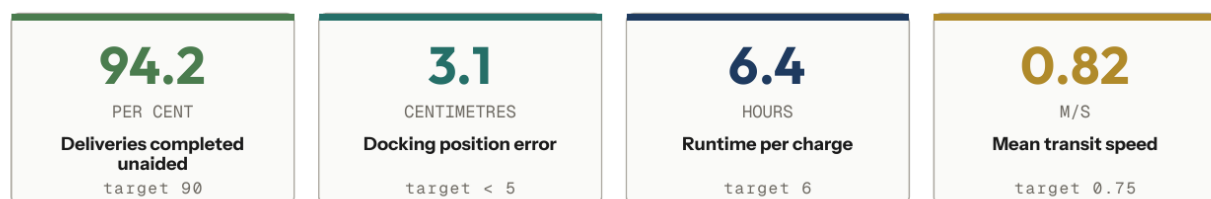
Chassis, drivetrain and castor		100%	TESTED
Power pack, BMS and E-stop loop		100%	TESTED
LiDAR mapping and localisation		100%	TESTED
Path planning and obstacle avoidance		90%	TUNING
Lift interface over BLE		70%	IN PROGRESS
Cargo lock, OTP and operator HMI		55%	IN PROGRESS

Figure 4 – What exists today. Grey portion is work still pending as on 5 September 2026.

Results. *Measured on the dates shown in Figure 5.*

What you measured	Target	Achieved
Deliveries completed unaided (120 runs)	$\geq 90\%$	94.2%
Docking position error	≤ 5 cm	3.1 cm
Runtime on one charge	≥ 6 h	6.4 h
Obstacle detection recall	$\geq 95\%$	97.6%
Build cost	$\leq ₹1,20,000$	₹1,04,600

MEASURED PERFORMANCE



TARGET VERSUS ACHIEVED

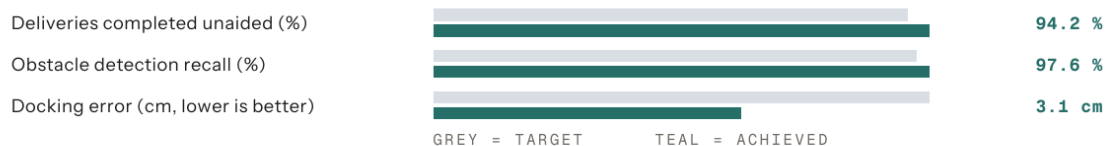


Figure 5 – Results of 120 supervised delivery runs across two floors between 12 and 30 August 2026.

H. WHO IT HELPS

The direct beneficiary is the campus administration, where two staff currently spend about 90 minutes a day on internal delivery trips. Because the map is learnt rather than programmed, the same robot can be moved to a new block by driving it around once. A hospital ward variant is the obvious extension, and the Bluetooth lift-calling method — which needs no change to the lift controller — is being taken up for a patent filing after the Expo.

I. REFERENCES

- [1] W. Hess, D. Kohler, H. Rapp and D. Andor, "Real-Time Loop Closure in 2D LIDAR SLAM," in Proc. IEEE ICRA, 2016, pp. 1271–1278.
- [2] D. Fox, W. Burgard and S. Thrun, "The Dynamic Window Approach to Collision Avoidance," IEEE Robotics & Automation Magazine, vol. 4, no. 1, pp. 23–33, 1997.
- [3] S. Macenski, T. Foote, B. Gerkey, C. Lalancette and W. Woodall, "Robot Operating System 2: Design, Architecture, and Uses in the Wild," Science Robotics, vol. 7, no. 66, 2022.
- [4] IS 15884 : 2010, Automated Guided Vehicles — Safety Requirements, Bureau of Indian Standards.

Signature of the Team Leader Date:	Signature of the Faculty Mentor Date:
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J. FOR OFFICE USE ONLY

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Evaluation Criterion	Max	Awarded	Remarks of the Jury
Problem and Objectives (Section E)	5		
Design and Technical Approach (Section F)	10		
Implementation and Results (Section G)	15		
Live Demonstration and Performance (Section H)	10		
Application Value (Section I)	5		
Presentation and Documentation (whole sheet)	5		
Total	50		

Screening decision	☐ Shortlisted ☐ Waitlisted ☐ Not shortlisted
Recommended for	☐ Award category: _____ ☐ Compendium ☐ IP support ☐ Incubation
Jury signature & date	